

Exploratory Data Analysis: Netflix Catalog Analysis

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1. Movie Runtime Evolution Across Decades (1960s – 2020s)

Does modern streaming favor shorter runtimes? We compute aggregate summary metrics across decades and test whether the compression between eras is statistically significant.

	decade	Count	Mean	Median	Std
0	1960	22	142.363636	141.5	40.359746
1	1970	63	119.111111	112.0	37.378669
2	1980	99	116.434343	104.0	35.695756
3	1990	194	113.917526	108.0	34.195921
4	2000	601	112.637271	108.0	28.312451
5	2010	3951	96.762086	96.0	25.961609
6	2020	423	89.522459	94.0	31.366490

-Hypothesis 1:

- + Null hypothesis (H0): The true mean movie runtimes of both decades are identical.
- + Alternative hypothesis (Ha): The true mean movie runtimes are different.

→ I used Welch's t-test with the 2000s and 2010s to find out whether the runtimes of both are different or not. With Welch's t-test (2000s vs 2010s): $t = 12.9432$, $p = 8.9068e-35$, there is enough evidence to reject the H0. Therefore, movie runtimes differ significantly across these decades.

-Hypothesis 2:

- + Null hypothesis (H0): The population distributions of movie durations across the three decades (2000s, 2010s, and 2020s) are identical.
- + Alternative hypothesis (Ha): At least one decade's movie duration distribution (or median) dominates/differs from the others.

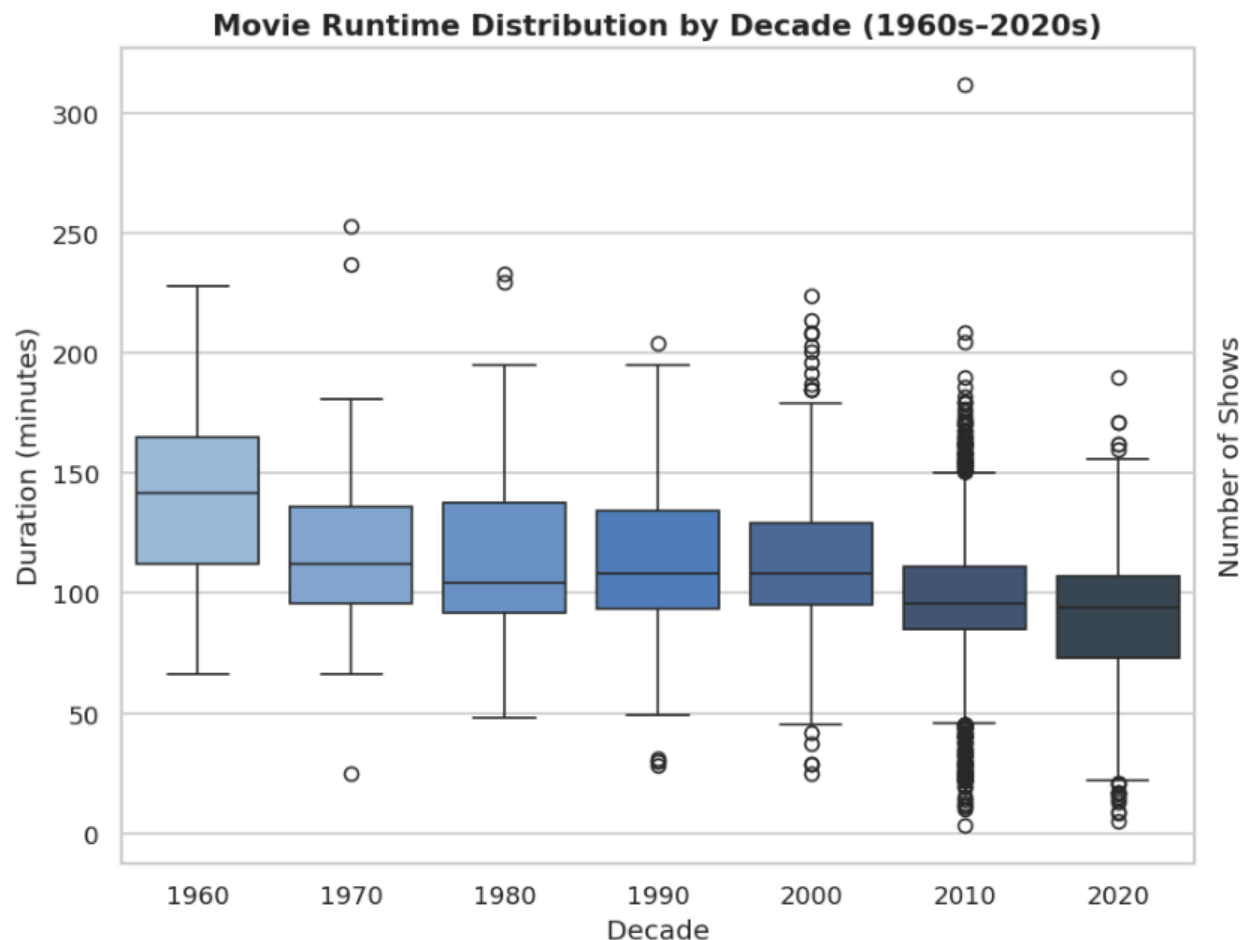
→ I applied Kruskal-Wallis test to the 2000s, 2010s, 2020s and got $H = 182.0766$, $p = 2.9011e-40$, which is drastically smaller than the standard significance threshold ($\alpha = 0.05$ or 0.01), the null hypothesis (H0) is rejected.

-Direction and Magnitude:

- + In the 2000s, the mean duration was 112.64 minutes (median: 108.00).
- + In the 2010s, the mean duration dropped to 96.76 minutes (median: 96.00), representing an average reduction of 15.88 minutes.
- + In the 2020s, the mean duration dropped further to 89.52 minutes (median: 94.00).

-Business Takeaway: Movie runtimes have undergone a statistically significant, structural compression in the streaming era rather than random catalog variation.

For easier visuality, I use box plot:



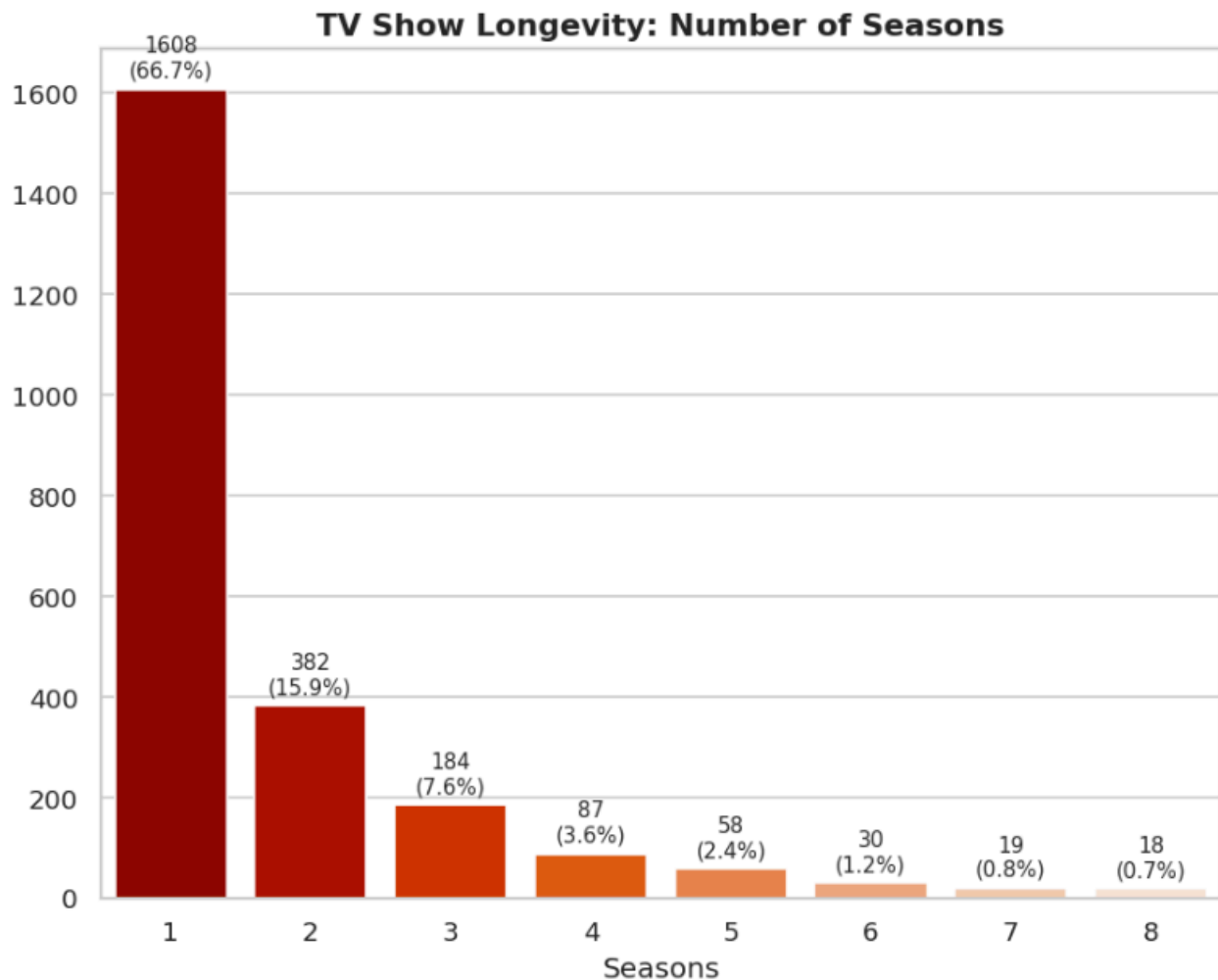
- Historical films from the 1960s had the highest median duration (around 140 minutes).
- Between the 1970s and 2000s, typical film durations remained relatively stable, with median runtimes hovering between 105 and 112 minutes.
- In the 2010s and 2020s, median durations noticeably compressed below the 100 minute mark (settling around 94 – 96 minutes).
- The 2010s cohort shows a dramatic surge in lower end outliers extending below 50 minutes (representing short films, anthology segments, and comedy specials) alongside high end outliers exceeding 200 to 300 minutes.
- This reflects a shift from rigid theatrical running times to streaming catalogs catering to diverse formats and shorter viewer attention spans.

2. TV Show Longevity & Season Drop-off Analysis

First, I count how many TV shows have exactly 1 season, 2 seasons, 3 seasons, and so on. Then, I create a DataFrame of total shows with the percentage of having one or more seasons.

duration	Total Shows	Percentage (%)
1	1608	66.72
2	382	15.85
3	184	7.63
4	87	3.61
5	58	2.41
6	30	1.24
7	19	0.79
8	18	0.75
9	8	0.33
10	6	0.25

For easier analysis, I use bar plots:



- Most TV series (66.7%) conclude after just 1 season.
- Only 15.9% (382 shows) reach Season 2, marking an immediate 76% drop-off in continuation after the debut season.
- Less than 15% of all shows survive past Season 2: Season 3 captures 7.6% (184 shows), Season 4 captures 3.6% (87 shows), and Season 5 captures 2.4% (58 shows).
- Extended franchises with 6 to 8+ seasons are rare outliers, making up roughly 1% or less each.

-Key takeaway: Television operates under an aggressive "test and abandon" model where series must achieve immediate viral performance in Season 1 or face cancellation, resulting in two-thirds of the catalog being single-season titles.

3. Multivariate OLS Regression: Predictors of Movie Duration

Here I apply a multivariate Ordinary Least Squares regression model in order to isolate the effect of time from confounding shifts in catalog composition. A downward trend in movie runtimes on a chart does not prove that movies themselves are becoming shorter because Netflix could simply be acquiring a higher proportion of short movies over time. Therefore, The regression model controls for genre and content maturity ratings simultaneously, making it possible to measure the true yearly change in film length while holding all other variables constant.

OLS Regression Results						
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Dep. Variable:	duration	R-squared:	0.269			
Model:	OLS	Adj. R-squared:	0.268			
Method:	Least Squares	F-statistic:	164.4			
Date:	Wed, 30 Sep 2026	Prob (F-statistic):	4.94e-324			
Time:	08:41:42	Log-Likelihood:	-22385.			
No. Observations:	4918	AIC:	4.479e+04			
Df Residuals:	4906	BIC:	4.487e+04			
Df Model:	11					
Covariance Type:	nonrobust					
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	coef	std err	t	P> t	[0.025	0.975]

Intercept	796.7920	71.165	11.196	0.000	657.276	936.308
C(rating)[T.PG-13]	2.5987	2.254	1.153	0.249	-1.821	7.018
C(rating)[T.R]	0.3710	2.164	0.171	0.864	-3.872	4.614
C(rating)[T.TV-14]	7.3293	2.050	3.576	0.000	3.311	11.347
C(rating)[T.TV-MA]	-3.6867	2.055	-1.794	0.073	-7.715	0.342
C(rating)[T.TV-PG]	-3.0760	2.063	-1.491	0.136	-7.120	0.968
release_year	-0.3503	0.035	-9.881	0.000	-0.420	-0.281
is_drama	18.4888	0.759	24.344	0.000	17.000	19.978
is_comedy	6.4535	0.802	8.050	0.000	4.882	8.025
is_action	12.9172	0.997	12.950	0.000	10.962	14.873
is_documentary	-8.9132	1.110	-8.029	0.000	-11.090	-6.737
is_children	-3.9840	1.940	-2.053	0.040	-7.788	-0.180
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Omnibus:	518.075	Durbin-Watson:	1.803			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	2260.689			
Skew:	0.437	Prob(JB):	0.00			
Kurtosis:	6.204	Cond. No.	4.37e+05			
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Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						
[2] The condition number is large, 4.37e+05. This might indicate that there are strong multicollinearity or other numerical problems.						

Structural explanations:

The empirical specification evaluates 4,918 feature films using ordinary least squares. The dependent variable is runtime in minutes (**duration**). Explanatory variables include continuous release year (**release_year**), binary indicators for key genres (drama, comedy, action, documentary, children), and categorical audience maturity ratings (**rating**) with the standard PG rating serving as the reference baseline.

Result interpretation:

- + The intercept is 796.79 ($p < 0.001$). This number represents the predicted runtime of a movie in year 0 AD with zero across all genre flags. As a result, the intercept is an uncentered baseline reference point rather than a realistic runtime.
- + R^2 of 0.269 with an F -statistic of 164.4 ($p < 0.001$), indicating that the selected predictors account for 26.9% of the overall variance in movie duration. The remaining ~73% of runtime variation is driven by unmodeled features: production budget, country of origin, director pacing, and theatrical release commitments.
- + The coefficient for release year is negative and statistically significant ($\beta = -0.3503, p < 0.001$). Holding genre and age rating constant, each additional calendar year reduces expected movie duration by roughly 0.35 minutes (approximately 21 seconds), which compounds to an estimated drop of 3.5 minutes per decade.
- + Genre indicators display substantial marginal effects: dramas and action titles extend runtime by 18.5 minutes and 12.9 minutes, respectively ($p < 0.001$).
- + Documentaries shorten runtime by 8.9 minutes ($p < 0.001$), while children's titles reduce it by 4.0 minutes ($p = 0.040$).
- + Most designations show no statistically significant difference relative to the PG baseline, with the exception of TV-14 titles, which average 7.3 minutes longer ($p < 0.001$).

Note clarification:

- + [2] appears because `release_year` is measured on an absolute scale (≈ 2000) while genre indicators are binary (0 or 1). So, centering `release_year` (e.g., `release_year - 2000`) removes the scale artifact without affecting coefficient slopes or p -values.